ML: MULTIPLE CHOICE QUESTIONS - 2

1. What is the main goal of Simple Linear Regression?
A. Classify data into categories
B. Predict a continuous outcome using one predictor
C. Reduce dimensionality
D. Cluster data

2. The equation of Simple Linear Regression is:
A. $y = mx^2 + c$
B. $y = mx + c$
C. $y = \log(x)$
D. $y = e^x$

3. In $y = mx + c$, what does 'm' represent?
A. Intercept
B. Error term
C. Slope
D. Mean

4. What does the intercept represent?
A. Value of y when x = 0
B. Value of x when y = 0
C. Maximum value
D. Variance

5. Which method is commonly used to estimate parameters?
A. Gradient Descent
B. Least Squares
C. K-Means
D. PCA

6. Multiple Linear Regression involves:
A. One predictor variable
B. Two or more predictor variables
C. Only categorical variables
D. No variables

7. General form of Multiple Linear Regression:
A. $y = mx + c$
B. $y = m_1x_1 + m_2x_2 + ... + c$
C. $y = x^2$
D. $y = \log(x)$

8. Multi-collinearity refers to:
A. High correlation between predictors
B. High correlation between output variables
C. Low variance
D. Non-linear relationship

9. Which metric evaluates model fit?
A. Accuracy
B. R-squared
C. Precision
D. Recall

10. Adjusted R^2 is used to:
A. Increase model complexity
B. Penalize unnecessary variables
C. Ignore variables
D. Measure error

11. Polynomial Regression is used when:
A. Data is linear
B. Data has a non-linear relationship
C. Data is categorical
D. Data is missing

12. A polynomial model of degree 2 looks like:
A. $y = mx + c$
B. $y = ax^2 + bx + c$
C. $y = \log(x)$
D. $y = e^x$

13. Increasing polynomial degree may lead to:
A. Underfitting
B. Overfitting
C. No change

D. Less variance

14. Polynomial regression is still:
A. Non-linear model
B. Linear model in parameters
C. Classification model
D. Clustering model

15. Which technique helps avoid overfitting?
A. Increasing features
B. Regularization
C. Ignoring data
D. Deleting model

16. Logistic Regression is used for:
A. Regression problems
B. Classification problems
C. Clustering
D. Reinforcement learning

17. The output of Logistic Regression is:
A. Continuous value
B. Probability between 0 and 1
C. Negative values only
D. Integers only

18. Which function is used in Logistic Regression?
A. Linear function
B. Sigmoid function
C. ReLU
D. Softmax only

19. Sigmoid function formula is:
A. $1 / (1 + e^{-x})$
B. x^2
C. log(x)
D. x + c

20. Decision boundary in Logistic Regression is typically:
A. Random
B. Fixed threshold (e.g., 0.5)
C. Infinite
D. Zero only

21. Which regression technique is best for binary classification?
A. Linear Regression
B. Logistic Regression
C. Polynomial Regression
D. Ridge Regression

22. Overfitting occurs when:
A. Model is too simple
B. Model fits noise in data
C. Model has low accuracy
D. Model has no variables

23. Underfitting occurs when:
A. Model is too complex
B. Model is too simple
C. Model fits perfectly
D. Model has high variance

24. Which term represents the error between predicted and actual values?
A. Residual
B. Gradient
C. Feature
D. Bias

25. Which algorithm uses a cost function like log loss?
A. Linear Regression
B. Logistic Regression
C. Polynomial Regression
D. KNN